Md Elius

PhD Candidate, Civil & Environmental Engineering · Atmospheric Dispersion Modeling & Emission Quantification
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PROFESSIONAL SUMMARY

Environmental engineer specializing in atmospheric dispersion modeling, Bayesian emission quantification, and monitoring network design for landfill methane and community air quality. Skilled in high-resolution and GPU-accelerated simulation (WRF-Chem, ERF, FastEddy) and experienced in turning large simulation and field datasets into defensible, decision-relevant findings for industry, regulatory, and academic partners.

EDUCATION

University of California, Berkeley — Berkeley, CA
PhD Candidate in Civil and Environmental Engineering · GPA 3.94 — *Expected May 2028*

University of Massachusetts Dartmouth — North Dartmouth, MA
MSc in Mechanical Engineering · GPA 3.97 — *Aug 2021 – Aug 2023*

Bangladesh University of Engineering & Technology (BUET) — Dhaka, Bangladesh
BSc in Mechanical Engineering · GPA 3.79 — *Feb 2015 – Apr 2019*

RESEARCH EXPERIENCE

Graduate Student Researcher — University of California, Berkeley — Berkeley, CA · Aug 2023 – Present

- Designed and evaluated fenceline and community-scale sensor networks for continuous emission monitoring using 30 m-resolution WRF-Chem large-eddy simulation (LES) plume ensembles over landfill sites.
- Developed a Bayesian MCMC inversion framework (Python, emcee) to quantify zone-level CH_4 emission rates across five landfill cover zones from fenceline synthetic sensor data.
- Performed site visits at two California landfills; analyzed mobile TCM tracer-release transects and walking SEM data to locate gas leaks and map spatial emission heterogeneity.
- Built Gaussian plume forward models (Pasquill-Gifford stability classes, Monin-Obukhov similarity meteorology, area-source dispersion) as a fast complement to WRF-based inversions.
- Validated model output against stationary, mobile, and drone observations using a Python/GIS toolkit (xarray, QGIS).
- Assessed community H_2S exposure: WRF-Chem analysis showed onsite sensors predict offsite exceedances; developing LSTM and gradient-boosting models to forecast concentrations at community receptors.
- Collaborate with industry and academic partners (Waste Management Inc., Project Canary, UC Davis, University of Delaware) on an EPA STAR-funded landfill methane monitoring program.

Research Assistant — University of Massachusetts Dartmouth — North Dartmouth, MA · Aug 2021 – Aug 2023

- Fabricated super-hydrophobic surfaces (aluminum, PDMS) by sandblasting, chemical etching, soft lithography, and spray coating.
- Characterized surfaces using scanning electron microscopy (SEM), water contact angle measurement, and total internal reflection; ran anti-biofouling tests in collaboration with biologists.
- Applied digital holographic microscopy (DHM) to track 3D bacterial motion and biofilm growth near surfaces.

Research Assistant — Bangladesh University of Engineering & Technology — Dhaka, Bangladesh · Apr 2018 – May 2019

- Developed a 2D ANSYS Fluent model using the SST k-ω turbulence model to predict shock-cell structure in an under-expanded jet from a converging nozzle.
- Characterized jet behavior by measuring static pressure and axial velocity at the nozzle exit across inlet pressures at constant back pressure, and analyzed the system's hysteresis.

AWARDS AND HONORS

- **EREF PhD Scholar (2025):** Environmental Research & Education Foundation PhD Scholarship for research on landfill methane emissions. Aug 2025 - Present
- **Dean's Scholarship:** awarded for GPA above 3.75 in each of four undergraduate years at BUET.

TEACHING AND SERVICE

Graduate Student Instructor — University of California, Berkeley Berkeley, CA · Fall 2025

- Led weekly discussion sections for a graduate-level Environmental Fluid Mechanics course, translating theory into worked problems for students from varied engineering backgrounds.
- Held office hours and designed and graded quizzes to assess student understanding.

Graduate Student Instructor — University of California, Berkeley Berkeley, CA · Spring 2024

- Led discussion sections and guided students in applying programming concepts (E7: Introduction to Computer Programming for Scientists and Engineers) and helped develop problem-solving exercises for course material.

Lecturer — Daffodil International University Dhaka, Bangladesh · Jan 2021 – May 2021

- Delivered lectures on core mechanical engineering concepts and mentored students on projects.

Member — UC Berkeley Environmental Engineering Seminar Committee Berkeley, CA

Treasurer— Bangladeshi Students and Scholars at Berkeley (BSSB) Berkeley, CA

PUBLICATIONS

- **M. Elius**, P. Yang, H. Gamarro, R. Yazdani, P. Imhoff, F.K. Chow. Performance evaluation and optimization of fenceline and community-scale sensor networks for continuous emission monitoring at municipal solid waste landfills. Under review, *Waste Management*.
- **M. Elius**, S. Richard, K. Boyle, W.S. Chang, P.H. Moisander, H. Ling. Impact of gas bubbles on bacterial adhesion on super-hydrophobic surfaces. *Results in Surfaces and Interfaces* 15, 100211 (2024).
- **M. Elius**, K. Boyle, W.S. Chang, P.H. Moisander, H. Ling. Comparison of 3D motion of bacteria with and without wall accumulation. *Physical Review E* 108(1), 014409 (2023).
- **M. Elius**, H. Ling. Effect of hologram plane position on particle tracking using digital holographic microscopy. *Applied Optics* 61, 9415–9422 (2022).

CONFERENCE PRESENTATIONS

- **M. Elius** (presenter), P. Yang, H. Gamarro, R. Yazdani, P. Imhoff, F.K. Chow. "Fixed point methane monitoring for landfill emissions detection." 10th Global Waste Management Symposium, Indian Wells, CA, Feb 2026.
- Imhoff, P. T., **Elius, M.**, Yang, K., Gamarro, H., Yazdani, R., Delkash, M., & Raaz, T. S. (2025). Digital Twins to Understand Methane Emissions From Landfills [Abstract GC23M-0856]. Presented at AGU Fall Meeting 2025.
- **M. Elius** (presenter), K. Boyle, P. Moisander, H. Ling. "Study of 3D bacterial motion and biofilm formation by digital holographic microscopy." APS Division of Fluid Dynamics 75th Annual Meeting, Indianapolis, IN, Nov 2022.
- **M. Elius** (presenter), M.M. Hasan, A.B.M.T. Hasan. "Underexpanded supersonic jet in imposed oscillating condition." 16th Asian Congress of Fluid Mechanics (16ACFM), Bengaluru, India, Dec 2019.

TECHNICAL SKILLS

Dispersion Modeling: WRF-Chem LES; FastEddy and ERF (GPU-accelerated computing)
Statistics & Inversion: Bayesian inversion; MCMC (emcee); uncertainty quantification; sensor network design and optimization
Machine Learning: Supervised and unsupervised learning; deep learning (CNN, LSTM, PINN; TensorFlow); genetic algorithms
Programming & Data: Python (NumPy, netCDF4, wrf-python, Matplotlib); MATLAB; Fortran; Bash; Git/GitHub; LaTeX
Geospatial & HPC: QGIS; Savio HPC cluster (Slurm job scheduling and workflow management)
CFD, FEA & CAD: ANSYS Fluent; COMSOL Multiphysics; SolidWorks (3D CAD and simulation)

REFERENCES

1. Fotini (Tina) Katopodes Chow
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   Professor of Civil & Environmental Engineering
   University of California Berkeley
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2. Hangjian Ling
   MS Advisor
   Associate Professor of Mechanical Engineering
   University of Massachusetts Dartmouth
   Email: hling1@umassd.ed